Hydrogen Lab Safety — User Manual

A guide to installing, using and maintaining the Hydrogen Lab Safety training platform.

Who this manual is for

Role	Who they are	What they can do
User	Any learner with an account	Read modules, explore the interactive lab, take the quiz, earn a certificate, give feedback
Admin	A user whose role was set to *admin*	Everything a user can, plus edit content and manage accounts from inside the website
Developer	Someone with access to the code and hosting	Install, configure, deploy and extend the platform

Each section below is split by role. Skip to the one that applies to you.

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1. Installation & Setup

Users

There is nothing to install. Open the website address your organisation gave you in any modern browser (Chrome, Edge, Firefox or Safari).

Create an account (first time only)

1. On the landing page, choose **Login**, then click **Create one**.
2. Fill in your details and choose your **Organisation**: *Fed Uni* or *Other*.
3. If you chose *Fed Uni*, enter your **Student ID** (required).
4. Click **Create account**. You will land on your dashboard.

Log in next time: go to **Login**, enter your email and password, and click **Sign in**.

Forgot your password? On the login page click **Forgot Password?**, enter your email, and follow the link in the reset email.

Admins

Admins use the same website as users. Two things to know first:

- **Becoming an admin.** Every new account starts as a regular user. An existing admin can promote you (see *Manage users* in section 2). The very first admin must be set up by a developer.
- **First-time content setup (one time only).** On a brand-new installation the content tables are empty, and the site shows built-in default content. To save that content so it can be edited and tracked, do the following **in this order**:
    a. Open each module, **/modules/hazards/1** through **/modules/hazards/5**. Switch on **Edit Mode** (top of page) and click **Save Changes** without changing anything.
    b. Open the lab at **Scenarios → Interactive Hydrogen Lab**. Switch on **Edit Mode** and click **Save Changes**.
    c. Go to **Quizzes** and click the 🖍️ **Edit** tab on the Hazards quiz card. Click **Save Changes**.

> *Do the modules first. The lab's hotspots link to modules, so saving the lab first will fail.*

These steps are technically optional: if you skip them, the site simply keeps showing the on-file defaults.

Developers

Requirements: Node.js 20.9 or higher, a free Firebase project and a free Supabase project.

1. **Install dependencies:** run `npm install` in the project folder.
2. **Firebase** (sign-in):
    - Create a project and register a web app.
    - Enable **Authentication → Sign-in method → Email/Password**.
    - Copy the web app config values from **Project Settings → Your apps**.
    - Generate a private key under **Project Settings → Service Accounts**.
3. **Supabase** (data and file storage):
    - Create a project.
    - In the **SQL Editor**, paste and run the whole of `supabase_setup.sql`. This creates all tables, storage buckets and permissions.
    - Copy the Project URL, the `sb_publishable` key and the `sb_secret` key from **Settings → API Keys**.
4. **Environment file:** create `.env.local` in the project root with the 12 Firebase and Supabase values listed in the README's *Environment Variables* table. Wrap only `FIREBASE_ADMIN_PRIVATE_KEY` in quotes.
5. **Allowed image domain:** in `next.config.js`, set the `hostname` to your Supabase project address (`yourprojectid.supabase.co`).
6. **Run:** `npm run dev`, then open http://localhost:3000.
7. **Create the first admin:** register an account on the site, then in Supabase open the `profiles` table and set that row's `role` to `admin`.
8. **Seed content:** ask the admin to follow the first-time content setup above, or do it yourself.
9. **Production:** `npm run build`, then `npm start`. When hosting, enter the same variables in your host's environment settings instead of uploading `.env.local`.
10. **Tests:** `npm test`. If any tests come back failed, then a mistake has been made during setup.

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2. Core Functionality Walkthrough

Users

Use the navigation bar at the top: **Home**, **Scenarios**, **Modules**, **Quizzes**, **About**, **Logout**.

Suggested learning path: Interactive Lab → Modules → Quiz → Certificate.
The interactive lab has links that take you to the modules. The quiz stays locked until you finish every module, so you need to complete them first.

Your dashboard

Click **Home**. You will see cards for Scenarios, Modules and Quizzes showing your progress, a certificate panel and a **Give Feedback** link.

Interactive lab

1. Click **Scenarios → Interactive Hydrogen Lab**.
2. Click the glowing hotspots on the lab image. Each opens a pop-up describing a hazard and what to do about it.
3. Click **Learn More →** in a pop-up (where shown) to jump to the related module. Some pop-ups also have a video.

Your first click on each hotspot is recorded on your dashboard.

Modules

1. Click **Modules → Hydrogen Safety Modules**. Each card shows a status (To do, In progress, Done) and a progress percentage. The filter bar at the top narrows the list by status.
2. Click a card to open the module. Read it by scrolling down through its numbered sections. Callouts marked 💡 highlight important points, and a **Key Takeaway** appears at the bottom.
3. Where a module has a video, click the video card to play it in a pop-up.
4. Use **← Previous** and **Next →** to move between modules, or **← Hazard Modules** to go back to the list.

How progress works: your progress is saved automatically as you scroll. A module counts as **Done** when you reach its last section.

- **Continue from saved progress** takes you back to where you left off.
- **Restart Module** (shown on a completed module) resets it to 0%. This also locks the quiz again until you finish the module once more.

Quiz

1. Click **Quizzes → Hydrogen Hazards Quiz**. If the card is greyed out, finish all modules first.
2. Answer every question. You can only submit once all are answered. Questions and answer order are random each attempt, and you may get a different set of questions each time.
3. Click **Submit**. Your score is shown as a percentage, and each missed question shows an explanation.
4. If you didn't pass, click **Retry Quiz** for a fresh set of questions.

Example: you answer 8 of 10 questions correctly, so your score is 80%. If the pass mark is 70%, you pass and a **Get Your Certificate →** button appears.

Leaderboard: after submitting, a banner asks whether to show your score. Choose ⏳ **Show My Score** to appear on the leaderboard, or 🔒 **Keep Private**. Your name only appears if you opt in. View it at **Quizzes → Student Leaderboard**.

> *Your most recent attempt replaces your previous result, including on the leaderboard, so a worse retry lowers your listed placement. The number of attempts is also used to determine placement.*

Certificate

Once all modules are Done and you've passed the quiz, click **Download Certificate →** on your dashboard (or **Get Your Certificate →** after passing). Check your name, then download the certificate as an image (PNG) to save or print. If you aren't eligible yet, the page tells you what's missing and links to it.

Feedback

Click **Give Feedback →** on the dashboard. Choose a star rating (1–5), a category, and write your message, then click **Submit**. All three are required.

Admins

Admins see everything users do, plus an **Administration** link in the navigation bar and **Edit Mode** tools on content pages. Regular users never see these.

Administration hub

Click **Administration** to find two tools: **User Management** and **Learner Feedback**.

Manage users

Open **Administration → User Management**.

- **Overview cards** show the number of users and administrators, and training completion.
- **Search** filters the list by email, name, organisation, role or user type.
- **Edit** (on a row) changes a user's **role** (user, staff or admin), **user type** and **organisation**. Click **Save** to apply. Email is read-only.
  Be careful not to accidentally change your own status from admin, as that will lock you out.
- **Progress** opens a read-only training record for that user: module-by-module progress, quiz score and certificate status.
- **Delete** removes a user, their progress and their login, after you confirm. You can't delete yourself.
- **Export Excel:** choose an organisation (*Fed Uni* or *Other*) and click **Export Excel**. You get a spreadsheet with Student ID, Name, Email, Score, Attempts, Passed and Last Attempted for everyone in that organisation. People who haven't attempted the quiz appear with 0 attempts.

View learner feedback

Open **Administration → Learner Feedback** for a summary (total responses, average rating, latest submission), a breakdown of star ratings, and the full list of submissions with rating, category, message, email and date.

Edit the interactive lab

1. Go to **Scenarios → Interactive Hydrogen Lab** and click the **Edit Mode** switch at the top-left. A banner confirms it is on.
2. **Move a hotspot:** drag it on the image. Hotspots turn blue in edit mode.
3. **Edit a hotspot:** click it (or pick it from the list) and change its title and description.
4. **Add or remove:** click **+** in the list header to add a hotspot. It appears in the centre of the image, ready to edit. Click ✕ next to a hotspot to delete it.
5. **Link to a module:** choose a topic, then a module under **Linked Module**. This creates the **Learn More** button. Choose *None* to remove the link.
6. **Video:** with a hotspot selected, use **Hotspot Video** to add a YouTube link or upload an mp4 file (max 50 MB). Videos save immediately.
7. **Change the background image:** click **Change Lab Image** and upload a replacement. This also saves immediately.

8. Click **Save Changes** to publish your hotspot edits, or **Reset to Defaults** to restore the original hotspots (this does not touch the lab image).

Edit a module

1. Open any module and click the **Edit Mode** switch at the top. The page above the editor previews your changes live.
2. **Module Details:** change the title, description, icon, icon background colour and badge number.
3. **Video:** add a YouTube link or an mp4 file (max 50 MB). Videos save immediately.
4. **Sections:** pick a section from the list to edit its heading, text, bulleted or numbered list, and callout. Use **+** to add, ↑ ↓ to reorder and ✕ to delete. Section numbers update automatically.
5. Click **Save Changes** to publish or **Reset to Defaults** to restore the original text.

Edit the quiz

1. Go to **Quizzes** and click the 🖍️ **Edit** tab on the quiz card.
2. At the top, set the **Title**, **Description**, **Pass Threshold** (0–100%) and **Pool Size** (how many questions each attempt draws; blank means all).
3. Pick a question from the list to change its text, options and explanation. Tick the radio button beside the correct option. Each question needs at least 2 options.
4. Tick **Core** to make a question appear in every attempt.
5. Use **+**, ↑ ↓ and ✕ to add, reorder and delete questions.
6. Click **Save Changes**. It stays disabled, with a warning, until any problems are fixed (for example an invalid pass mark, or a pool size larger than the number of questions).

Good to know

- **Unsaved-changes warning:** if you try to close the tab or click a link with unsaved edits, the site asks you to confirm. Turning Edit Mode off does *not* discard your changes.
- **The quiz lock applies to everyone**, including admins. Adding a new module raises the requirement for all users, including those who had already finished.
- **Videos** save on their own, separately from **Save Changes**.

Developers

Developers can do the following (see the project's `README.md`, `EDITING_GUIDE.md` and `ADDITIONAL_INFO.md` for details):

- Change the built-in default content: modules, lab hotspots and the quiz question bank (requires redeployment).
- Add a new module topic or a standalone page using the provided templates.
- Edit content tables directly in the Supabase dashboard.
- Change the look and feel: colours, fonts and page layouts.
- Manage hosting, environment variables and deployments.
- Run the automated tests and the continuous-integration workflow.
- Fix known issues listed in `BUG_REPORT.md`.
- Set or change the first admin account directly in the database.

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3. Troubleshooting

Users

Problem	Solution
First click on **Login** after logging out does nothing	Click **Login** a second time. This is a known quirk.
The Quiz card is greyed out, or I'm sent back to the Quizzes page	You haven't finished every module yet. Complete them all under **Modules**.
A module still shows In progress	Scroll all the way to the last section so it registers as complete.
My quiz is locked again	You used **Restart Module**. Finish that module again.
No certificate available	You need all modules Done and a quiz pass (about 70% or more). The certificate page tells you which part is missing.
My leaderboard placement went down	Your latest attempt replaces the earlier one. Retake the quiz to improve it. The number of times a quiz has been attempted also affects your placement.
I can't change my leaderboard choice	The choice only appears right after you submit a quiz.
I can't submit feedback	Make sure you are logged in and that rating, category and message are all filled in.
Forgot password	Use **Forgot Password?** on the login page.
Page won't load or looks wrong	Refresh the page, and check that you are logged in.

Admins

Problem	Solution
I can't see Administration or Edit Mode	Your account isn't an admin. Ask a developer or existing admin to change your role.
Saving a user's **user type** fails with an error	The dropdown currently includes options the database rejects. Use *public*, *lecturer* or *researcher*, and ask a developer to fix the list.
A user is missing from the Excel export	Exports include only users whose organisation is exactly *Fed Uni* or *Other*. Check or correct it with **Edit**.
Student ID is blank in the export	Student ID is only collected at sign-up for Fed Uni users, and it can't be edited here.
A user's Student ID or organisation is blank from the start	Sign-up occasionally saves an incomplete profile. Set the organisation with **Edit**. Student ID needs a developer.
Save Changes is greyed out in Edit Mode or the Quiz Editor	Read the warning on the page. Common causes: a hotspot linked to a topic but no module, a quiz question with fewer than 2 options, or an invalid pass threshold or pool size.
Saving the lab fails on a new installation	Save every module first (see *First-time content setup*).
A module isn't in the hotspot's Linked Module list	It must be saved at least once in Edit Mode first.
Quiz won't unlock for a learner	They haven't completed every module, or they restarted one. Check their record using **Progress**.
Progress numbers or Certificate status look inconsistent	The admin screens only count the Hazards modules, and some figures are calculated differently from the learner's certificate page. Treat the learner's own certificate page as the final word.
A deleted user's feedback still appears	Feedback is not removed when a user is deleted.
My edits disappeared	You left the page without saving, or clicked **Reset to Defaults**. Always click **Save Changes**.
Uploaded video is rejected	Use a YouTube link or an mp4 file under 50 MB.
A new lab image doesn't show	Refresh the page.

Developers

Problem	Solution
`npm install` or `npm run dev` fails	Confirm Node.js is 20.9 or higher.
Sign-in or the whole app errors on start	Check `.env.local`: all 12 values present, no quotes or spaces except the private key, then restart the dev server.
`permission denied for table …`	Re-run the permissions section of `supabase_setup.sql`. Tables need explicit grants.
Saving the lab fails with a foreign-key error	Seed modules before hotspots.
Images from storage don't load	Check the hostname in `next.config.js` matches your Supabase project.
Private-key errors	Make sure `FIREBASE_ADMIN_PRIVATE_KEY` is wrapped in quotes with literal `\n` line breaks.
Admin tools say access denied	Confirm the user's `role` is `admin` in the `profiles` table.
Changes to default content don't appear	Default content is built into the code and needs a redeploy. Live content is edited in the app or in Supabase.

For anything not covered here, check `BUG_REPORT.md` for known issues.